

[Assignment-1]

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Section: 05

Course : CSE 330
code

Faculty : [AGU]
Initial

Answer to the question no 1

a

Here, $\beta = 2$

$m = 4$

and $-4 \leq e \leq 2$

The maximum numbers:

Standard Form: $+(0.1111)_2 \times 2^2$

IEEE Denormalized Form: $+(1.1111)_2 \times 2^2$

IEEE Normalized Form: $+(0.11111)_2 \times 2^2$

b

Non-negative minimum numbers:

Standard Form: $+(0.1000)_2 \times 2^{-4}$

IEEE Denormalized Form: $+(1.0000)_2 \times 2^{-4}$

IEEE Normalized Form: $+(0.10000)_2 \times 2^{-4}$

C

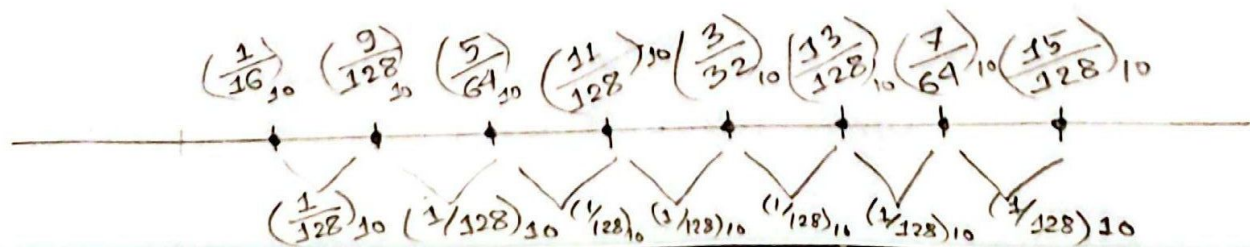
Standard Form : $F = \pm (0.d_1 d_2 d_3 \dots d_m)_{\beta} \beta^e$
 $\beta = 2, m = 4, -4 \leq e \leq 2$

For, $e = -3, m = 4$ and $\beta = 2$,

The combinations are,

$$\text{Total} = 2^3 = 8 \Rightarrow \begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{array}$$

$$\begin{aligned} (0.1000)_2 \times 2^{-3} &= (2^{-1} \times 1) \times 2^{-3} = (0.0625)_{10} = \left(\frac{1}{16}\right)_{10} \\ (0.1001)_2 \times 2^{-3} &= (2^{-1} + 2^{-4}) \times 2^{-3} = (0.070)_{10} = \left(\frac{9}{128}\right)_{10} \\ (0.1010)_2 \times 2^{-3} &= (2^{-1} + 2^{-3}) \times 2^{-3} = (0.078)_{10} = \left(\frac{5}{64}\right)_{10} \\ (0.1011)_2 \times 2^{-3} &= (2^{-1} + 2^{-3} + 2^{-4}) \times 2^{-3} = (0.0859)_{10} = \left(\frac{11}{128}\right)_{10} \\ (0.1100)_2 \times 2^{-3} &= (2^{-1} + 2^{-2}) \times 2^{-3} = (0.09375)_{10} = \left(\frac{3}{32}\right)_{10} \\ (0.1110)_2 \times 2^{-3} &= (2^{-1} + 2^{-2} + 2^{-3}) \times 2^{-3} = (0.109)_{10} = \left(\frac{7}{64}\right)_{10} \\ (0.1101)_2 \times 2^{-3} &= (2^{-1} + 2^{-2} + 2^{-4}) \times 2^{-3} = (0.101562)_{10} = \left(\frac{13}{128}\right)_{10} \\ (0.1111)_2 \times 2^{-3} &= (2^{-1} + 2^{-2} + 2^{-3} + 2^{-4}) \times 2^{-3} = (0.117)_{10} = \left(\frac{15}{128}\right)_{10} \end{aligned}$$



So, The number line is equally spaced.
The difference between two point is

$$\frac{1}{128}$$

Ans to the question no 2

a

Let,

$$a = (2.23)_{10}$$

$$\begin{array}{r} 2 \overline{) 2} \\ \underline{2} \\ 0 \end{array} \quad \begin{array}{r} 2 \overline{) 1} \\ \underline{2} \\ 0 \end{array} \quad \begin{array}{r} 0 \\ 1 \end{array} \quad \uparrow$$

$$\begin{array}{l} 0.23 \times 2 = 0.46 \quad \text{---} \quad 0 \\ 0.46 \times 2 = 0.92 \quad \text{---} \quad 0 \\ 0.92 \times 2 = 1.84 \quad \text{---} \quad 1 \\ 0.84 \times 2 = 1.68 \quad \text{---} \quad 1 \\ 0.68 \times 2 = 1.36 \quad \text{---} \quad 1 \\ 0.36 \times 2 = 0.72 \quad \text{---} \quad 0 \\ 0.72 \times 2 = 1.44 \quad \text{---} \quad 1 \\ \vdots \end{array}$$

$$a = (2.23)_{10} = (10.0011101 \dots)_{2}$$

Here, $\beta = 2$, $m = 3$, $e_{\min} = -2$, $e_{\max} = 2$

So,

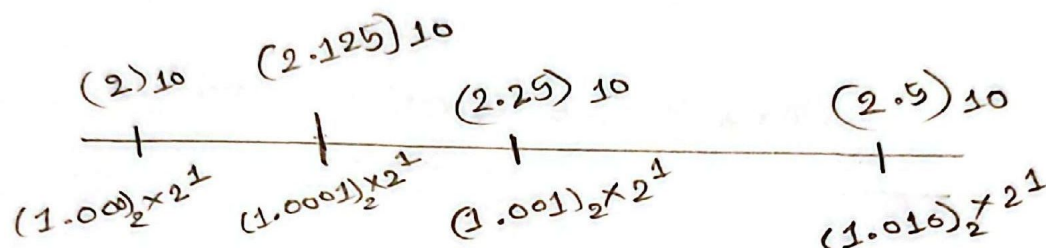
$$\begin{aligned} a &= (10.00110\dots)_2 \times 2^0 \\ &= (1.000110\dots)_2 \times 2^1 \end{aligned}$$

So,

$$(1.000)_2 \times 2^1 = (1 \times 2^0) = (2)_{10}$$

$$(1.001)_2 \times 2^1 = (1 \times 2^0 + 2^{-3}) \times 2 = (2.25)_{10}$$

$$(1.010)_2 \times 2^1 = (1 \times 2^0 + 2^{-2}) \times 2 = (2.5)_{10}$$



$a = (2.23)_{10}$ is more close to $(2.25)_{10}$

~~or $(1.001)_2 \times 2^1$~~ . So, we can round off it to $(1.001)_2 \times 2^1$.

$$\therefore fl(a) = (1.001)_2 \times 2^1$$

again,

$$\text{let, } b = (2.2018)_{10}$$

$$\begin{array}{r} 2 \overline{) 2} \\ \underline{2} \\ 0 \\ \underline{0} \\ 0 \\ \underline{0} \\ 0 \end{array} \quad \begin{array}{c} 0 \\ \uparrow \\ 1 \end{array}$$

$$\therefore b = (2.2018)_{10}$$

$$= (10.001100\dots)_2$$

$$= (10.001100\dots)_2 \times 2^0$$

$$= (1.0001100\dots)_2 \times 2^1$$

$$\begin{array}{rcl} 0.2018 \times 2 & = & 0.4036 \quad \text{---} 0 \\ 0.4036 \times 2 & = & 0.8072 \quad \text{---} 0 \\ 0.8072 \times 2 & = & 1.6144 \quad \text{---} 1 \\ 0.6144 \times 2 & = & 1.2288 \quad \text{---} 1 \\ 0.2288 \times 2 & = & 0.4576 \quad \text{---} 0 \\ 0.4576 \times 2 & = & 0.9152 \quad \text{---} 0 \\ & & \vdots \end{array}$$

$b = (2.2018)_{10}$ is more close to $(2.25)_{10}$

So, we can round off it to $(1.001)_2 \times 2^1$

$$\therefore fl(b) = (1.001)_2 \times 2^1$$

b

$$a = (2.23)_{10} = (10.0011101 \dots)_2$$

$$fl(a) = (2.25)_{10}$$

$$b = (2.2018)_{10} = (10.001100 \dots)_2$$

$$fl(b) = (2.25)_{10}$$

Now, for a,

$$\begin{aligned} \text{Rounding error} &= |fl(a) - a| \\ &= |2.25 - 2.23| \end{aligned}$$

$$= (0.02)_{10}$$

$$= (0.0000010 \dots)_2$$

for b

$$\begin{aligned} \text{Rounding error} &= |fl(b) - b| \\ &= |2.25 - 2.2018| \\ &= (0.0482)_{10} \\ &= (0.000011 \dots)_2 \end{aligned}$$

$$\begin{array}{l} 0.02 \times 2 = 0.04 \rightarrow 0 \\ 0.04 \times 2 = 0.08 \rightarrow 0 \\ 0.08 \times 2 = 0.16 \rightarrow 0 \\ 0.16 \times 2 = 0.32 \rightarrow 0 \\ 0.32 \times 2 = 0.64 \rightarrow 0 \\ 0.64 \times 2 = 1.28 \rightarrow 1 \\ 0.28 \times 2 = 0.56 \rightarrow 0 \\ \vdots \end{array}$$

$$\begin{array}{l} 0.0482 \times 2 = 0.0964 \rightarrow 0 \\ 0.0964 \times 2 = 0.1928 \rightarrow 0 \\ 0.1928 \times 2 = 0.3856 \rightarrow 0 \\ 0.3856 \times 2 = 0.7712 \rightarrow 0 \\ 0.7712 \times 2 = 1.5424 \rightarrow 1 \\ 0.5424 \times 2 = 1.0848 \rightarrow 1 \end{array}$$

Q

Let, $a = (2.23)_{10}$

$$= (10.0011101 \dots)_2 \times 2^0$$

$$= (1.00011101 \dots)_2 \times 2^1$$

$$= (0.100011101 \dots)_2 \times 2^2 \quad [\text{IEEE Normalized Form}]$$

Given,

(before round off)

$$e_{\min} = -2 \quad \text{and} \quad e_{\max} = 2$$

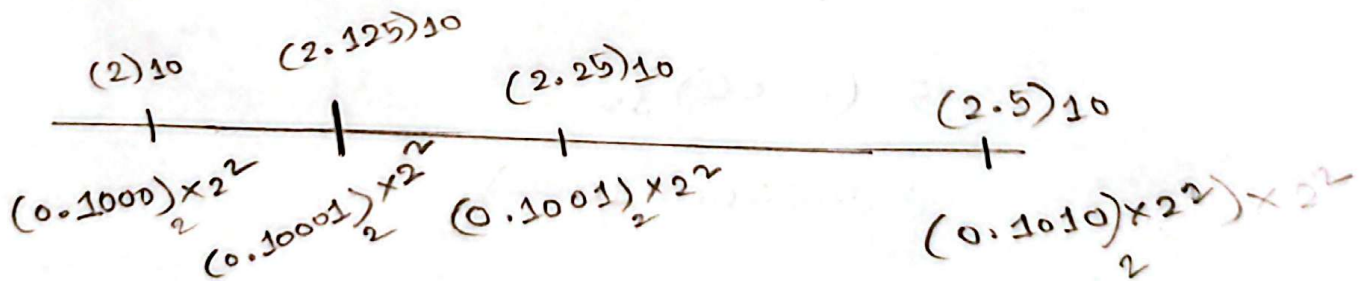
From the normalized form we can see that e is in the range. So, it can be represented in normalized form.

So,

$$(0.1000) \times 2^2 = (2)_{10}$$

$$(0.1001) \times 2^2 = (2.25)_{10}$$

$$(0.1010) \times 2^2 = (2.5)_{10}$$



$a = (2.23)_{10}$ is more close to $(2.25)_{10}$.

So, we can round off it to ~~$(1.001)_2 \times 2^2$~~

$$(0.1001) \times 2^2$$

$$\therefore S1(a) = (0.1001) \times 2^2$$

Let,

$$b = (2.2018)_{10}$$

$$= (10.001100\dots)_2$$

$$= (10.001100\dots)_2 \times 2^0$$

$$= (1.0001100\dots)_2 \times 2^1$$

$$= (0.10001100\dots)_2 \times 2^2$$

[IEEE Normalized Form] (before round off)

From the normalized form we can see that e is in the range. So, it can be represented in normalized form.

Here,

$$b = (2.2018)_{10} \text{ is more close to } (2.25)_{10}$$

and greater than middle point, so,

we can round off it to $(0.1001)_2 \times 2^2$.

$$\therefore fl(b) = (0.1001)_2 \times 2^2$$

— x —